

MARTIN BISSONNETTE

Nanotechnology Engineering Student · University of Waterloo

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SKILLS

Lab Experience

Photolithography, Wet/Dry Etching, Cleanroom Protocols, Organic Chemistry, Optical Alignment, Laser Operation & Safety

Software & Simulation

SolidWorks, AutoCAD, Fusion, FEMM, KLayout, KiCAD, Gwydion, ImageJ/Fiji

Programming

Python, MATLAB, Excel VBA, C++, HTML, CSS

Lab Equipment

AFM, SEM, 2-Photon DLW, XRD, PECVD, RIE, Microscopy, Spectroscopy (UV-Vis, FTIR, Raman, Ellipsometer), TMA, TGA, DSC

Hardware

Microcontrollers (Arduino, STM32, ESP32, GD32F2), Raspberry Pi, CNC systems

Languages

Fluent in English and French (DELF Certified)

WORK EXPERIENCE

Irradiant Technologies

Materials & Packaging Engineer

September 2025 – May 2026

Boston, Massachusetts

- Developed semiconductor packaging methods for proprietary nano-porous scaffold materials, increasing quality, yield and throughput by 10x and enabling the company to acquire a \$500K contract with Coherent.
- Advanced the development of high-fidelity 3D micro-electronic systems under DARPA's AMME program, supporting proposals that secured over \$4M in contract awards;
- Conducted several AFM and SEM measurement cycles at Harvard CNS (Trained User) that directly contributed to a \$1.8M milestone payment and multiple customer developments.
- Designed and produced photosensitive silver reduction materials for use in multi-photon 3D lithography systems, optimizing for conductivity, surface quality, and sub-micron resolution
- Implemented a multi-FOV stitching feature for the point-scan lithography system, enabling printing of large-scale conductive structures at nanoscale resolution — a capability that allowed for the creation of micro-electronic and GHz RF structures for DARPA grant proposals.
- Assisted in the design and validation of an optical measurement system for insertion loss and beam quality of printed photonic waveguides, enabling data-driven optimization of print geometry and process conditions.
- Co-authored a low-cost open-access two-photon lithography system for academic research, published on ResearchGate. ([link](#))

C2MI

Assembly Process Intern

January 2025 – May 2025

Bromont, Quebec

- Performed microfabrication and testing in a Class 1000 cleanroom using semiconductor and optical packaging tools.
- Performed R&D on photonic wire bonds as an alternative photonic interconnect method in collaboration with IBM; designed a photonic waveguide chip using KLayout for testing insertion losses of optical interconnects
- Developed a laser-induced graphene (LIG) process for PCB manufacturing, enabling flexible, high-conductivity electroplating on sub-mm traces comparable to bulk electrodes..
- Documented and tested a wafer-level packaging method to embed chips into remolded wafers, enabling further processing and higher package density.
- Characterized materials and assembly quality using microscopy, spectroscopy, and electro-mechanical analysis techniques.

SMART LAB

Part-Time Research Assistant

May 2025 – August 2025

Waterloo, Ontario

- Synthesized hydrogels and conducted quantitative material characterization to evaluate and tailor properties for targeted drug delivery.

PROJECTS

UW Silica Integrated Optics — Co-founder

December 2023 – Ongoing

- Established an official University design team for Nanotechnology Engineering students; led fabrication of a high-speed programmable laser shutter and actively created DOEs to test laser parameters for voxel creation.

EDUCATION

University of Waterloo

Bachelor of Applied Sciences in Honours Nanotechnology Engineering 82% Avg.

Fall 2023 – Spring 2028

Ontario, Canada